

Data Science — Course Syllabus

Learn Python, Machine Learning, Pandas, NumPy and real data science projects.

Duration	6 Months	Level	Advanced
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Tools & Technologies Covered

Python, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn

Course Modules

Module 1: Python for Data Science

- NumPy for numerical computing
- Pandas for data wrangling
- Data visualization with Matplotlib and Seaborn

Module 2: Statistics & Probability

- Descriptive and inferential statistics
- Probability distributions
- Hypothesis testing

Module 3: Machine Learning

- Regression and classification algorithms
- Model evaluation metrics
- Feature engineering and selection with Scikit-learn

Module 4: Advanced Topics

- Clustering and dimensionality reduction
- Introduction to deep learning concepts
- Working with real-world messy datasets

Module 5: Capstone Project

- End-to-end data science project from raw data to model
- Project presentation and documentation

Learning Outcomes

- Perform statistical analysis and build machine learning models
- Use Python's data science stack (NumPy, Pandas, Scikit-learn) confidently
- Deliver a complete, presentable data science project

For more details, batch timings and enrollment, visit the [EduClasses website](#) or contact our admissions team.